Anshita Bhardwaj

Machine Learning Engineer
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Education

Arizona State University — Tempe, AZ
Master of Science in Data Science, Analytics and Engineering — GPA: 3.8/4.0
Courses: Machine Learning, Data Processing at Scale, Statistical Modeling, Data Mining

Amity University — Jaipur, Rajasthan
Bachelor of Technology in Computer Science and Engineering (Minor in Management) — GPA: 3.9/4.0
Courses: Data Structures and Algorithms, Object-Oriented Programming, Databases, Operating Systems

Technical Skills

Languages: Python, Java, C#, C/C++, SQL, JavaScript, TypeScript, Go, R, Bash

AI & Generative AI: LLM agents with tool use, RAG, prompt engineering, LangGraph, LangChain, semantic search (FAISS), hallucination guardrails, human-in-the-loop, model evaluation

Machine Learning & Statistics: Supervised and unsupervised learning, XGBoost, scikit-learn, SHAP, Fairlearn, causal inference and regression discontinuity, time-series forecasting (LSTM), BERT, hypothesis testing

Backend & Full Stack: FastAPI, Flask, REST API design, microservices, integration testing, React, Next.js, TypeScript, GraphQL

Data Engineering: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Snowflake, dbt, Kafka, Spark, ETL pipelines, Pandas, NumPy, query optimization

Cloud, MLOps & Tools: AWS, Google Cloud Platform (Vertex AI), Docker, Kubernetes, Terraform, MLflow, CI/CD (GitHub Actions), Claude Code, Cursor, Gemini, Tableau, Power BI

Work Experience

Arizona State University | *Research Assistant* — Tempe, AZ · *Feb 2024 – Present*

- Built automated Python pipelines (Pandas, NumPy, SciPy) over **1.5M+ US birth records spanning 28 years**, raising data granularity **200x** for a study of how **federal disaster declarations** affect US birth rates; earned a **10,000 USD scholarship**.
- Designed a **temporal regression discontinuity (RDD)** study on an **age-standardized general fertility rate**, estimating heterogeneous effects by **hazard type** and **socioeconomic status**; caught a composition artifact in my own denominator, corrected it with **ASFR weighting**, and **re-ran the full study** in one pass.

Escorts Kubota Limited | *Data Analyst* — Haryana, India · *Jun 2023 – Jun 2024*

- Cut inventory costs **10%** and raised planning efficiency **25%** across **50+ products** with **LSTM demand-forecasting models** and dashboards adopted daily by Material Planning and Warehouse Operations.
- Improved forecast accuracy **18%** with feature engineering pipelines, and reduced equipment downtime **30%** with a separate **predictive-maintenance model**.

JustDial Ltd. | *Data Analyst Intern* — Maharashtra, India · *Jun 2022 – Aug 2022*

- Fine-tuned a **BERT** model to **92% accuracy** on **10,000+ reviews** and built a **TF-IDF retrieval pipeline** that raised search relevance **15%**.
- Wrote and optimized **advanced SQL** against **Microsoft SQL Server** and built web scraping systems feeding downstream search and NLP pipelines.

Projects

TalentLens: Agentic AI Platform with Enterprise RAG | *Python, LLM agents, RAG, FastAPI, React, Kubernetes*

- Built and shipped a **five-subsystem platform** end to end as sole engineer: **LLM agents with tool use** and RAG over a 4-layer PostgreSQL and dbt warehouse of **43,844 records**, with REST boundaries, **Kubernetes** autoscaling, and MLflow model tracking.
- Added **human-in-the-loop controls**, drift detection and bias auditing, capturing **180+ reviewer overrides** as labeled training examples.

Responsible AI for Healthcare Risk Prediction | *XGBoost, SHAP, Fairlearn, scikit-learn, Streamlit*

- Built an end-to-end **predictive model** reaching **94% accuracy** on hospital readmission risk across multi-year patient data.
- Cut **model bias 40%** with **SHAP** explainability and **Fairlearn** auditing, delivered as a **Streamlit** dashboard for clinical review.

Docu-Sync: AI Agent for the Development Workflow | *Python, RAG, Gemini, Cursor, GitHub* | *IBM Hackathon*

- Built an agent that keeps documentation in step with code through a **4-stage multimodal RAG pipeline**, opening a **human-approved pull request** rather than committing directly.
- Added an **evaluation suite across 11 dimensions** (semantic accuracy, retrieval use, hallucination, confidence calibration) and a confidence score on every update so the reviewer knows how far to trust it.

Publications

Bhardwaj, A., et al. (2025). *Quantifying hallucination bias in AI-generated deepfakes.* Research Square, 2025 (Preprint).

Bhardwaj, A., Tirunagari, P., & Vashishta, N. (2023). *Unleashing the power of LSTM networks for sentiment analysis in Twitter.* AIAMMS 2023.

Vashishta, N., Bhardwaj, A., & Tirunagari, P. (2023). *Reimagining CAPTCHA security.* AIAMMS 2023.